

Consistency-Constrained Multi-Task Bengali Hate Speech Detection with Faithfulness-Evaluated Generative Explanations

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Abstract—While Multi-Task Learning (MTL) has advanced Bengali hate speech detection by jointly predicting dimensions like hate type, target, and severity, current architectures suffer from severe logical inconsistency. Independent classification heads frequently output contradictory predictions (e.g., predicting “No Hate” while simultaneously identifying a “Target”). Furthermore, while recent frameworks attempt generative explainability, the faithfulness of these free-text rationales to the model’s internal reasoning remains untested in Bengali. In this paper, we propose the first Consistency-Constrained MTL framework for Bengali hate speech. By integrating a novel contradiction-penalty loss during training, we significantly reduce logical inference errors. Additionally, we introduce a generative rationale head trained on silver-standard explanations and present the first rigorous evaluation of generative faithfulness (Comprehensiveness and Sufficiency) in a low-resource language.

Index Terms—Hate Speech Detection, Bengali NLP, Multi-Task Learning, Explainable AI, Logical Consistency, Faithfulness

I. INTRODUCTION

The proliferation of digital hate speech on social media platforms necessitates robust automated moderation systems, particularly for widely spoken but historically under-resourced languages like Bengali [1], [2]. Early efforts in Bengali hate speech detection primarily focused on simple binary classification (i.e., distinguishing between hateful and non-hateful text) using traditional machine learning and early deep learning architectures [3], [4]. However, as the socio-political dynamics of online harassment have evolved, platforms require multidimensional insights—such as identifying the specific type of hate (e.g., religious, political), the targeted demographic, and the severity of the toxicity [5].

This demand catalyzed a shift toward Multi-Task Learning (MTL) paradigms, prominently featured in the recent BLP-2025 shared task [6], [7]. By leveraging shared transformer encoders, MTL architectures attempt to jointly model these related dimensions, achieving high empirical accuracy [8], [9]. Despite these advancements, current state-of-the-art Bengali models suffer from two critical vulnerabilities: a **Consistency Gap** and a **Faithfulness Gap**.

The Consistency Gap arises because task-specific classification heads are typically optimized independently during

inference. Consequently, models frequently output logically impossible label combinations—such as classifying a comment as “Non-Hate” while simultaneously predicting it targets a “Religious Minority” with “Severe” intensity [10], [11]. The Faithfulness Gap refers to the severe lack of algorithmic transparency. While recent datasets like BanHADEX [12] allow models to generate human-readable explanations for their predictions, it remains entirely unknown whether these generated rationales are “faithful” to the model’s actual internal reasoning, or if the model is merely hallucinating plausible text [13], [14].

To address these vulnerabilities, this paper proposes a novel, joint-optimization architecture. Our specific contributions are threefold:

- 1) We introduce a **Consistency Penalty Loss** that natively embeds cross-task logical constraints into the objective function, explicitly penalizing contradictory multi-task predictions.
- 2) We integrate an **Explanation-Guided Generative Head**, utilizing a few-shot LLM pipeline to construct silver-standard rationales that align massive multi-task datasets with explainability corpora.
- 3) We present the first application of perturbation-based faithfulness metrics—Comprehensiveness, Sufficiency, and Area Over the Perturbation Curve (AOPC)—to generative Bengali hate speech models.

II. RELATED WORK

A. Multi-Task Bengali Hate Speech Detection

The shift to fine-grained analysis was heavily catalyzed by the BanglaMultiHate dataset, which annotated over 50,000 comments across Hate Type, Target, and Severity [5], [15]. This led directly to the BLP-2025 shared task [6]. Top-performing systems unanimously abandoned single-task pipelines in favor of MTL architectures or hierarchical ensembles [16], [17].

The winning system (Code_Gen) utilized a heavy ensemble of BanglaBERT and MuRIL combined with Token-Aware

Adversarial Contrastive Training to achieve state-of-the-art F1 scores [8]. Similarly, HateSense applied Focal Loss and Odds Ratio Preference Optimization (ORPO) to handle severe class imbalance [9]. Other notable approaches include hierarchical multiclass classification by PerceptionLab [18], ensemble learning by CUET_Sntx_Srfs [19], and joint MTL training by Retriv [20]. Additionally, datasets like BIDWESH have highlighted the necessity of maintaining accuracy across regional Bengali dialects [21].

However, these models share a fundamental flaw: their task-specific heads operate independently during inference. Because they do not learn the strict, hierarchical logical rules of the annotation schema, they are prone to logical contradictions. This lack of joint constraint formulation is a known failure point in complex MTL environments [22].

B. Logical Consistency Constraints

In broader NLP and Computer Vision, the problem of contradictory predictions in multi-task setups is actively researched. When tasks are optimized independently, the overall belief system becomes incoherent [10], [23]. Researchers have explored cross-task consistency learning frameworks that mathematically penalize incompatible outputs during training [11], [24].

For instance, constrained multi-task learning has been successfully applied to bridging resolution [24] and vision-language models [23] by injecting probabilistic compatibility matrices. Despite the proven efficacy of consistency-aware fine-tuning, no prior work has successfully adapted model-level consistency constraints for Bengali hate speech, leaving current architectures vulnerable to contradictory label generation.

C. Generative XAI and Faithfulness

The demand for trustworthy AI has driven research into Explainable AI (XAI) [25], [26]. Early Bengali explainers, such as DeepHateExplainer [27], utilized post-hoc methods like Layer-wise Relevance Propagation (LRP) for token-level highlighting. However, highlighting explicit slurs is insufficient for understanding implicit hate or sarcasm.

Recent high-resource frameworks have advanced explainability significantly. UNIREX proposed a unified learning framework for language models encompassing both rationale extraction and classification [28]. HARE introduced step-by-step reasoning for explainable hate speech detection in English [29]. Furthermore, the HAXE framework introduced attention-rationale alignment to force transformer weights to align with human rationales [30]. Similarly, X-MuTeST began exploring faithfulness metrics for other Indic languages like Hindi and Telugu [31].

For Bengali, the BanHADEX dataset marked a critical shift by introducing human-annotated, free-text rationales [12], [32]. This enabled generative explainability via sequence-to-sequence architectures. However, a critical "faithfulness gap" remains. The plausibility-faithfulness trade-off dictates that highly fluent rationales often do not accurately reflect the

model's true reasoning [14], [33], [34]. To date, ERASER-style metrics (Comprehensiveness and Sufficiency) [13] and Area Over the Perturbation Curve (AOPC) [35] have not been applied to verify the faithfulness of Bengali generative hate speech models, risking the deployment of models that hallucinate justifications.

III. METHODOLOGY

To bridge the identified gaps, we construct an integrated architecture that fuses a shared transformer encoder with discriminative multi-task heads and a dedicated sequence-to-sequence generation head.

A. Dataset Alignment via Silver-Standard Rationalization

The primary barrier to unified MTL and generative XAI is a taxonomic misalignment: the BanglaMultiHate dataset possesses multi-task labels (Type, Target, Severity) but lacks explanations, while the BanHADEX dataset possesses explanations but lacks the same dimensional granularity. We address this using Silver-Standard Rationalization. We deploy a frontier Large Language Model (e.g., Gemini 1.5 Pro) in a few-shot capacity to generate synthetic, structured explanations for a subset of BanglaMultiHate. These rationales are explicitly conditioned on the ground-truth labels.

B. Explanation-Guided Joint Architecture

Using a lightweight open-weight LLM (e.g., Llama-3 8B) adapted for Bengali via LoRA, or a standard encoder like BanglaBERT [36], the architecture is branched:

- **Discriminative Heads:** Three independent linear layers map the pooled contextual embeddings to Hate Type, Target, and Severity logits.
- **Generative Head:** An autoregressive decoder generates the explanation tokens conditioned on the input context.

C. Joint Optimization & Consistency Penalty

The training objective jointly optimizes the cross-entropy losses of the multi-task classification heads alongside the autoregressive next-token prediction loss, modified by a novel Consistency Penalty. The unified loss function is formulated as:

$$\mathcal{L} = \alpha\mathcal{L}_{type} + \beta\mathcal{L}_{target} + \gamma\mathcal{L}_{sev} + \delta\mathcal{L}_{gen} + \lambda\mathcal{L}_{consist} \quad (1)$$

The discriminative heads ($\mathcal{L}_{type}, \mathcal{L}_{target}, \mathcal{L}_{sev}$) utilize Focal Loss to manage extreme class imbalances (e.g., the scarcity of "Severe" labels). $\mathcal{L}_{gen}$ applies standard causal language modeling loss over the explanation tokens. Crucially, $\mathcal{L}_{consist}$ is a compatibility matrix penalty that scales exponentially when the model predicts logically contradictory parameters (e.g., $P(\text{Type} = \text{None}) \times P(\text{Severity} = \text{Severe})$), guaranteeing coherent outputs.

D. Perturbation-Based Faithfulness Evaluation

We evaluate the generative head using Comprehensiveness and Sufficiency. By applying token perturbation (masking generated rationale tokens from the original input), we calculate the Area Over the Perturbation Curve (AOPC). This mathematically verifies that the generative explanation serves as a faithful window into the latent feature space governing the classification.

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